

What Reprices and How Fast in Weighted-Average-Maturity Pass-Through

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Abstract

Weighted average maturity is a natural first statistic for judging how quickly a rate shock reaches a sovereign interest bill. It is also easy to overuse. Maturity describes when debt leaves the stock; not all repricing waits for that event. This paper separates the Portuguese debt stock into maturity-driven, reset-driven, and retail-flow components and asks how much of the opening stock has actually received a new rate by each horizon. For Portugal, a 100 basis-point shock reprices 10.90 percentage points more of the stock within one year than the usual weighted-average-maturity proxy implies, worth about EUR 300 million in first-year interest. The reason is mostly composition: 14.2 percent of the stock is floating-rate or inflation-linked and reprices within a year without maturing. The paper also tests whether retail savings certificates add a behavioural channel. The answer is cautious. The competing-return spread is not estimated precisely, the bootstrap interval for asymmetry is $[-0.020, +0.044]$, and the kernel does not beat the maturity benchmark out of sample. The raw episode is nevertheless informative, because households moved money into certificates during the 2022–2023 tightening rather than redeeming them. The result is a practical measurement correction, not a claim to forecast interest expenditure better from a short annual sample.

1 Introduction

When interest rates rise, the public debate usually moves quickly from the market yield to the budget line. That leap hides the part of the problem that matters most for annual fiscal accounts: the old debt stock does not wake up one morning carrying the new rate. Some of it matures and is refinanced; some of it resets while staying in place; some of it is retail money that may flow in or out when the offered return changes. Weighted average maturity is a useful summary of the first channel, but it is often asked to do the work of all three.

The shortcut is attractive because it is clean. A portfolio with average maturity m can be treated as if each remaining euro has a constant probability $1/m$ of repricing every year. The companion paper uses exactly that device in its refinancing section because it makes the timing assumption visible. This paper keeps the same question and removes the homogeneity assumption. The object of interest is not the average date at which a bond matures; it is the share of the opening stock that has actually received a new rate by each horizon.

That distinction changes the interpretation of a rate shock before any behavioural story enters. Floating-rate and inflation-linked liabilities do not need to leave the stock in order to reprice. Retail instruments can also add new money at the prevailing rate, which is economically similar to repricing from the budget's point of view even when it is not redemption and replacement. A maturity-calibrated hazard cannot see either channel. It treats repricing and retirement as the same event, which is why it has trouble with portfolios that contain meaningful reset and retail blocks.

The Portuguese case is useful because those blocks are observable enough to separate. IGCP publishes monthly instrument-class stocks and portfolio indicators, and the 2022–2023 tightening

created a natural stress test for retail savings certificates. The raw pattern is already informative: households moved money into certificates when short rates rose, because the certificate remuneration formula became attractive relative to bank deposits. The behavioural channel therefore runs through subscriptions, not through the redemption margin that a simple “holders flee to higher rates” story would suggest.

The main result is a measurement correction rather than a forecast. For Portugal, a 100 basis-point shock reprices 10.90 percentage points more of the stock within one year than the weighted-average-maturity proxy implies, worth about EUR 300 million in first-year interest. Most of the credibility of that number comes from composition, not from a fragile behavioural regression: 14.2 percent of the stock is floating-rate or inflation-linked and reprices within a year regardless of residual maturity. The behavioural estimates are deliberately treated with caution. The spread coefficient is imprecise, the bootstrap interval for asymmetry covers zero, and the backtest does not establish a forecasting gain over the maturity benchmark.

The paper’s practical message is correspondingly modest. Weighted average maturity should not be discarded; it should be used for the part of the stock whose repricing is genuinely tied to maturity. Reset instruments and retail flows need their own clocks. Once those clocks are separated, the fiscal burden also needs an explicit GDP path, because the same euro interest bill can look quite different as a share of GDP.

2 Related literature

The paper sits between debt-maturity theory, sovereign pass-through accounting, and household-deposit pass-through. The maturity literature explains why the term structure of public liabilities matters for fiscal risk. Missale and Blanchard [8] make debt maturity central to the burden of government debt. Angeletos [3] shows how a non-contingent maturity structure can replicate state-contingent fiscal payoffs in a complete set of maturities. Greenwood, Hanson and Stein [7] study maturity choice as a comparative-advantage problem for government debt management. Arellano and Ramanarayanan [4] and Broner, Lorenzoni and Schmukler [5] show, in different settings, that maturity is not a neutral technical detail: it interacts with spreads, crisis conditions, and rollover risk.

This paper does not estimate an optimal maturity structure and does not model default. Its narrower concern is the measurement step that often comes before those questions. If a sensitivity exercise says that a rate shock reaches the budget according to weighted average maturity, it has already assumed that the portfolio is homogeneous enough for one maturity clock to stand in for all repricing clocks. The Portuguese data give a concrete way to check that assumption.

The retail part of the design is closer to the deposit-rate literature than to standard sovereign bond pricing. Drechsler, Savov and Schnabl [6] emphasise that monetary tightening changes the relative return on deposits and therefore the behaviour of deposit holders and banks. Portuguese savings certificates are not bank deposits, but they compete with household deposit rates and are held by households. That makes the subscription margin a plausible place to look for state-dependent pass-through, while also making clear why the paper cannot infer household-level behaviour from aggregate stock data.

The accounting side links the paper back to the companion burden study. Escolano [9] gives the standard debt-dynamics bridge from interest rates and growth to public-debt ratios, while Blanchard [2] shows why that bridge became politically important in a low-rate environment. This paper is about the step before that bridge is used: how much of the inherited stock has actually received the new rate. That step is easy to compress into one maturity statistic, but the compression is exactly where the Portuguese portfolio loses information.

3 Institutional setting

Portugal is an unusually good laboratory for a behavioural repricing test because a large, identifiable share of its debt is held in retail savings instruments — Certificados de Aforro and Certificados do Tesouro — that households may redeem on demand subject to lock-ups and penalties. As of June 2026 these accounted for 15.4 percent of State direct debt. A holder facing a change in the competing return on bank deposits can act on it, so the repricing of that block is in principle a behavioural object rather than a contractual one.

The 2022–2023 tightening provides a clean episode in the everyday sense, not in the econometric sense. Certificados de Aforro grew from EUR 12.9 billion in June 2022 to EUR 32.6 billion by May 2023, a rise of 151 percent, peaking at a monthly inflow of EUR 3,549 million. That is a large movement in a visible instrument class, but it is also one episode, taking place while inflation, household liquidity, deposit competition, and policy communication were all moving.

The direction of the movement is the useful clue. A simple behavioural story would say that tightening speeds pass-through because retail holders redeem low-yielding instruments and reinvest elsewhere at higher rates. Portugal does not fit that story. The certificate’s remuneration formula tracked short rates closely enough that the instrument became attractive relative to deposits and drew money in. For the budget, money arriving at the prevailing rate is still repriced money. The channel exists, but it is not the one implied by a redemption hazard.

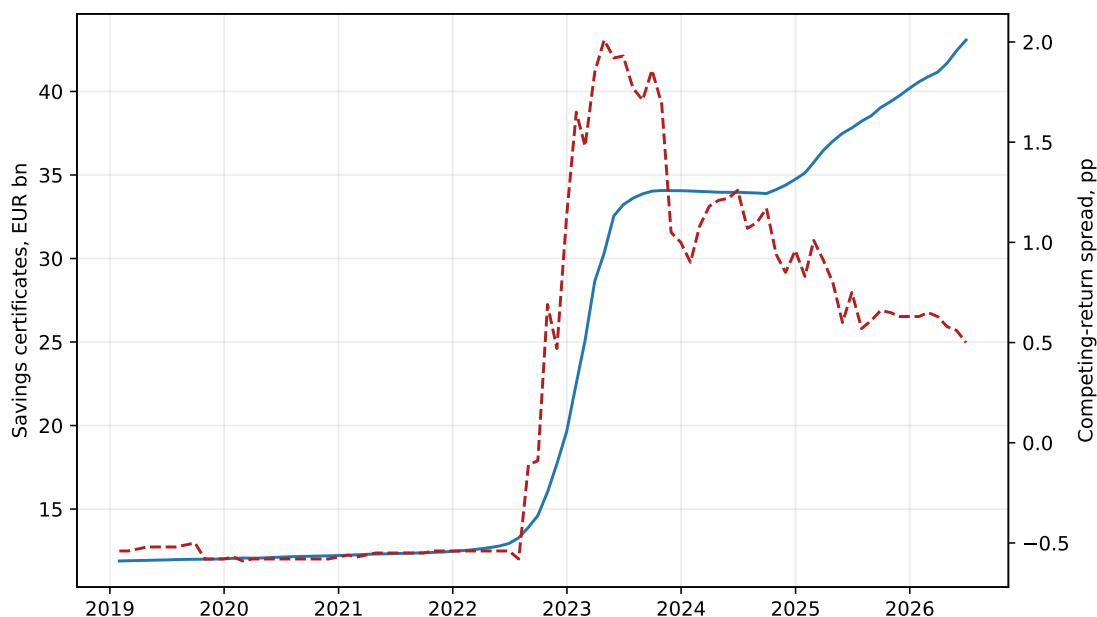


Figure 1: Retail savings certificates and the competing-return spread. The stock is the outstanding amount of Certificados de Aforro; the spread is the policy-rate measure used in the panel less the Portuguese household deposit-rate measure.

Figure 1 is included because the shape of the episode is easier to understand visually than in coefficient form. The stock rises during tightening, then the change in terms offered on new subscriptions arrests the inflow. The figure is not causal evidence on its own, but it is enough to discipline the empirical design: the paper should model the subscription margin and should not pretend to have observed gross redemptions.

4 Data construction

Instrument-class stocks and portfolio indicators come from IGCP historical series, monthly. Competing-return covariates come from the ECB Data Portal. The measurement layer for aggregate debt, GDP, and interest is imported from the companion paper rather than recomputed. The data architecture is intentionally plain: one monthly panel for the repricing question, one annual aggregate dataset for the burden translation, and generated manuscript tables so the paper does not contain hand-typed results. Table 1 summarises the portfolio state used to initialise the kernel. The average maturity input is the IGCP annual-report statistic [10].

Table 1: Portfolio inputs for the repricing kernel	
Quantity	Value
Observation month	June 2026
Average residual maturity (years)	7.52
Fixed-rate share (%)	85.8
Floating or indexed share (%)	14.2
Retail certificate share (%)	15.4
Panel start	February 2001
Panel end	June 2026

Source: author calculations from IGCP monthly stocks and portfolio indicators.

The data are informative, but they do not do more than they say. The unit of observation is a class-month, not a holder. Retail certificate data are published only as aggregate stocks, so the panel cannot separate households that subscribe, redeem, roll over, or remain inactive within the month. The paper therefore avoids individual language. It is a portfolio-flow design, not a household microeconomic design.

The other constraint is that only net stock is observed. Gross subscriptions and gross redemptions are not published in the source series. A flat month is consistent with no activity and with large offsetting flows, which means a redemption hazard is not identified. The empirical design estimates the subscription margin, where positive net flow bounds the repriced amount from below. Months of net outflow are recorded as outflows, not converted into zero repricing events.

The contractual side has a different limitation. No dated redemption schedule is published in machine-readable form for the whole stock, so the contractual track uses a stylised retirement profile with the same mean maturity. That is not a reconstruction of IGCP’s cash-flow calendar. It is a deliberately transparent counterfactual that lets the paper isolate the composition problem: what changes when part of the stock reprices without maturing?

5 Kernel framework

Let $K(h)$ be the share of the stock outstanding at the horizon origin that carries a repriced rate by horizon h . The notation is simple, but the interpretation is important. A euro can join $K(h)$ because it matures and is reissued, because it resets while staying outstanding, or because new retail money arrives on the offered rate. Those are different clocks. Putting them into one hazard is convenient, but it hides the part of the mechanism this paper is trying to measure.

The contractual component is debt that reprices by maturing and being reissued. This is the component for which a maturity clock is conceptually appropriate. If a redemption schedule is known, it should be used directly. If only a mean maturity is known, a stylised profile is a second-best approximation. The reset component is different. Floating-rate and inflation-linked debt can receive a new rate without leaving the stock, so it is not a survival event. The behavioural

component is different again: retail money can arrive on the prevailing rate, and that arrival can depend on the shock. The empirical test below is deliberately modest; it asks only whether the subscription margin carries a detectable response to the competing-return spread.

The benchmark against which we measure is the standard proxy,

$$K^{\text{WAM}}(h) = 1 - e^{-h/m}, \quad (1)$$

with m the published weighted average maturity. Equation (1) is memoryless: it implies a long right tail, leaving a substantial share unrepriced after a decade.

The estimated kernel is

$$K(h, \Delta) = K^C(h) + K^R(h) + K^B(h, \Delta), \quad (2)$$

where K^C is contractual repricing, K^R is reset repricing, and K^B is behavioural repricing under shock Δ . The bias relative to the maturity proxy is then

$$B(h, \Delta) = K(h, \Delta) - K^{\text{WAM}}(h). \quad (3)$$

The decomposition reported in the tables splits B into a shape component and a behavioural component. The shape component is identified from portfolio composition and the stylised retirement track; the behavioural component uses the estimated subscription response.

6 Estimation

The specification was frozen before fitting and estimated once; no specification search was performed. The outcome is the repriced share of the opening stock. The regressors are the signed parts of the lagged competing-return spread, an indicator for the June 2023 change in the terms offered on new subscriptions, and portfolio composition. Standard errors are Newey–West. With two instrument classes, cluster-robust inference would carry a false sense of authority, so it is not reported.

The result is a null. The spread-widening coefficient is +0.0187 with a standard error of 0.0148 ($p = 0.20$). The asymmetry the design turns on — whether widening and narrowing load differently — has a bootstrap interval of $[-0.020, +0.044]$ across 1,000 moving-block replicates and covers zero.

A falsification test supports reading this as imprecision rather than contamination: the fixed-rate share, a portfolio statistic no household observes, loads at -0.00019 ($p = 0.86$) and leaves the other coefficients essentially unchanged.

That null is not surprising. The sample contains one large widening episode, so its timing cannot be separated cleanly from everything else moving with it. The certificate’s contractual remuneration rate is not published machine-readably, so a short-rate index stands in and attenuation of unknown size is expected. The outcome is also a one-sided bound: positive net flow gives a lower bound on newly repriced money, while net-outflow months tell us that money left but not how much gross subscription and redemption occurred underneath. A more flexible specification would make the table look more elaborate without creating the missing gross-flow data.

Table 2: Subscription-margin regression

Term	Coefficient	Newey–West s.e.	<i>p</i> -value
const	-0.0107	0.0181	0.56
spread_widening_pp	0.0187	0.0148	0.20
spread_narrowing_pp	0.0077	0.0060	0.20
post_policy_break	-0.0213	0.0160	0.18
average_residual_term_years	0.0024	0.0029	0.41

Outcome: repriced lower-bound share of the opening class stock. Source: author calculations from IGCP and ECB data.

Table 2 is deliberately shown in full because a compressed “behavioural response” number would overstate what the data can support. The policy break has the expected negative sign, yet it is not precise enough to carry a structural interpretation. The maturity covariate is similarly weak. The result is still useful, but mainly as a restraint. The behavioural component can be carried through the kernel as a sensitivity, not as the centrepiece of the paper.

7 The kernel and the bias

Table 3 reports the bias at a 100 basis-point shock. The sign is positive at every reported horizon: *the standard proxy understates how much of the stock reprices*. The table also shows why that statement should not be reduced to “WAM is too slow”. The bias is a combination of an early reset effect, a stylised contractual-shape effect, and a behavioural component whose central value is reported but whose uncertainty matters.

Table 3: Bias in the weighted-average-maturity proxy, +100 bps

Horizon	Proxy share (%)	Estimated share (%)	Total bias (pp)	Shape (pp)	Behaviour (pp)
1	12.45	23.35	10.90	7.44	3.46
3	32.90	41.68	8.78	-1.59	10.37
5	48.57	53.00	4.43	-5.85	10.28
10	73.55	76.41	2.86	-2.30	5.16

Source: author calculations. Shape and behaviour sum to the total.

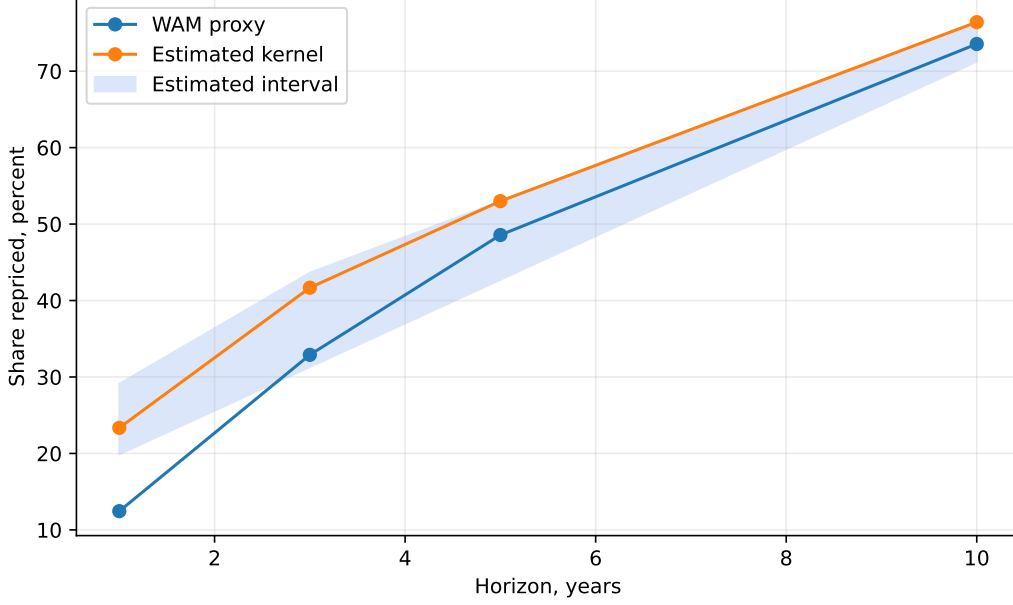


Figure 2: Weighted-average-maturity proxy and estimated repricing kernel. The interval is the generated behavioural interval around the estimated kernel.

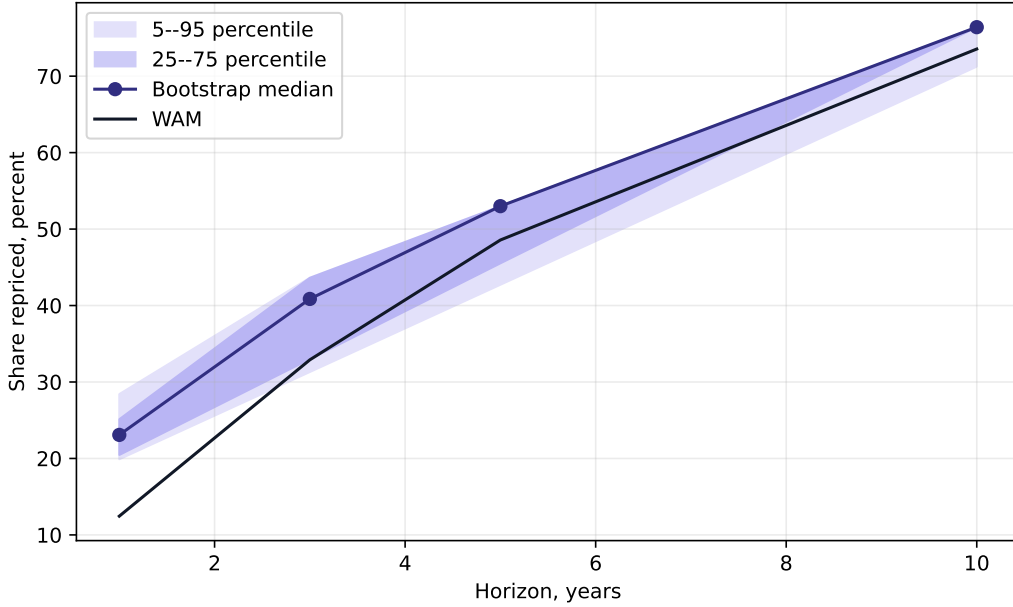


Figure 3: Bootstrap uncertainty in the estimated repricing kernel. The bands propagate the moving-block bootstrap draws for the behavioural response through the full kernel.

The mechanism is compositional. At one year the shape component contributes 7.44 percentage points and requires no behavioural assumption whatsoever: 14.2 percent of the stock reprices within a year on its own reset cycle, and Equation (1) cannot represent it.

Two qualifications. First, the shape term is *not monotone* in the horizon: it turns negative by three and five years, because beyond the reset window the memoryless tail retires more than a profile that has already retired its fast component. “The geometric kernel is too slow” is true early and false later. Second, the behavioural component’s band runs from zero at every horizon, because the estimate behind it is the null of Section 6. We report the central path only with

that band; a point estimate without it would be precisely the failure this paper criticises.

In fiscal terms the one-year bias is 0.098 percent of GDP, about EUR 300 million.

Figure 2 makes the non-monotonicity visible. The estimated kernel is above the WAM proxy at the reported horizons, but the gap does not widen smoothly. The reset block brings pass-through forward; the stylised contractual profile can then sit below the memoryless tail at middle horizons. That pattern is the reason for using the phrase “compositionally biased” rather than the easier phrase “too slow”. The standard proxy is not wrong in one direction at every point of the curve; it is using the wrong clock for part of the stock.

Figure 3 adds the empirical uncertainty that the point curve alone hides. The band is not mainly a story about debt managers changing contractual maturities; those inputs are held fixed. It is the uncertainty from the retail-flow estimate carried through the kernel. That is why the band is useful even though the behavioural coefficient is a null. It shows which part of the measurement correction is stable composition and which part is still a thin-data behavioural estimate.

8 Pass-through

Simulating forward under the estimated kernel also clarifies the denominator problem. The companion paper holds nominal GDP fixed across the ten-year horizon to isolate refinancing arithmetic. That convention is transparent, but it is not innocuous. At a 100 basis-point shock and a five-year horizon, the incremental burden is 0.475 percent of GDP under the frozen-GDP assumption and 0.391 percent under central nominal growth. Realistic growth removes about 18 percent of the measured effect.

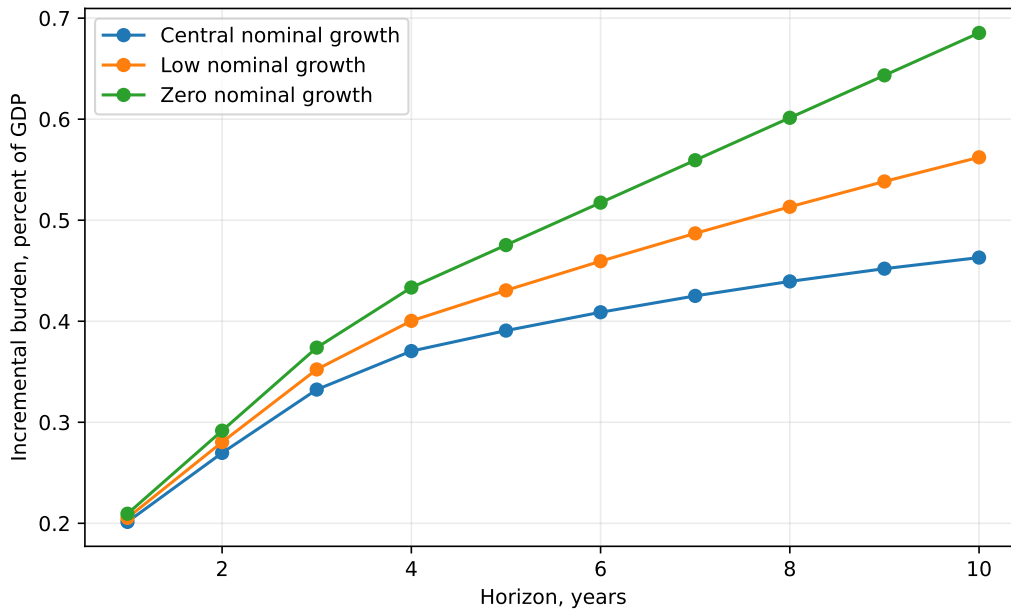


Figure 4: Incremental burden under a +100 basis-point shock across nominal-GDP paths. The lines differ only in the denominator path.

Table 4: Half-life of incremental pass-through under central nominal growth

Shock (bps)	Maximum burden effect (% GDP)	Half-life (years)
-200	0.864	3.0
-100	0.432	3.0
-50	0.216	3.0
50	0.232	2.0
100	0.463	2.0
200	0.926	2.0

The half-life is the first horizon at which the absolute burden effect reaches half of its simulated maximum.

We note one artefact so it is not mistaken for a finding. Simulated half-lives differ by shock direction, but that asymmetry is manufactured by clipping the behavioural response at zero, which prevents a rate fall producing a negative contribution. It is a modelling choice, and the estimated asymmetry was a null.

The pass-through exercise therefore has two separate lessons. The kernel decides how much of the stock has received the shock. The GDP path decides how large that same euro interest change looks as a burden ratio. A fiscal note that freezes GDP is useful for comparability with full-pass-through arithmetic, but it should not be read as a central projection. The economic object is a euro interest bill; the political and fiscal object is usually the bill as a share of GDP. The two should not be allowed to drift into each other.

9 Uncertainty and sensitivity

The deterministic paths above are useful because they isolate one mechanism at a time. They are not enough by themselves. A reader also needs to know whether the result survives plausible movements in the modelling choices. The paper therefore adds two generated checks: a Monte Carlo fan chart for the burden path and a one-at-a-time sensitivity table for the kernel assumptions.

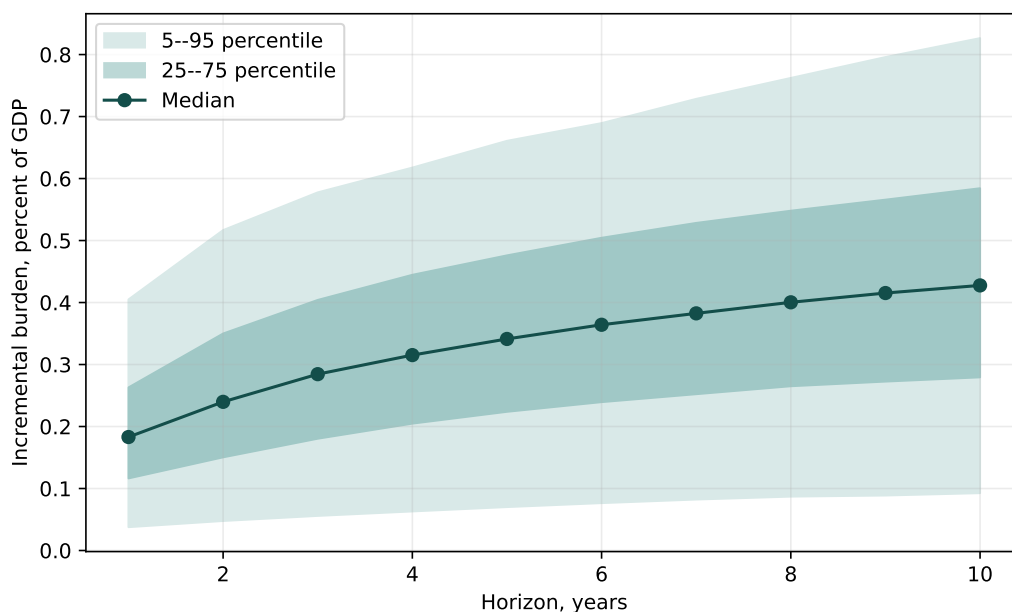


Figure 5: Monte Carlo fan chart for incremental burden pass-through. The draws combine rate-shock uncertainty, nominal-GDP-growth uncertainty, and bootstrap uncertainty in the behavioural response.

Figure 5 uses 2,000 generated draws. The mean shock in the draw set is 99 basis points and the mean nominal-growth draw is 4.0 percent. By the fifth year, the simulated burden effect has a median of 0.341 percent of GDP, with a 5–95 percentile interval from 0.070 to 0.661. The wide band is not a failure of the kernel; it is the fiscal reality that the same repricing path can look quite different once the shock size and GDP denominator are allowed to vary.

Table 5: Sensitivity of the kernel to modelling assumptions, +100 bps

Scenario	One-year bias (pp)	Five-year repriced share (%)	Five-year burden (% GDP)
Central kernel	10.90	53.00	0.457
Slow reset	3.81	53.00	0.457
Fast reset	10.90	53.00	0.457
Memoryless contractual shape	14.99	63.79	0.550
Behaviour off	7.44	42.72	0.368
Behaviour lower bound	7.44	42.72	0.368
Behaviour upper bound	16.65	53.00	0.457

The table varies one assumption at a time around the central kernel. Source: author simulations.

Table 5 asks a different question. Instead of randomising the macro environment, it moves the main modelling assumptions one at a time. The one-year bias range across those checks is 12.85 percentage points. The slow-reset row matters most because it directly attacks the compositional claim: if reset instruments are assumed to take longer to receive the new rate, the first-year correction shrinks. The behavioural rows matter less for the central conclusion. They widen or narrow the total, but they do not remove the reason the WAM proxy is incomplete.

10 Implications for fiscal reporting

The easiest way to use this result is not to replace weighted average maturity with a new black box. The useful change is smaller and more practical: report a short repricing bridge beside the usual maturity statistic. The bridge should state how much of the opening stock is fixed-

rate and maturity-driven, how much is reset-linked, and how much belongs to retail or other instruments whose stock can move with offered terms. A reader can then see why the same published maturity can imply different first-year budget exposure across countries or across years.

This also changes how uncertainty should be presented. A conventional sensitivity table often gives a clean interest-cost number for a rate shock and then leaves the timing assumption implicit. In the Portuguese case the more honest presentation would separate the contractual component, which is mostly a portfolio-accounting issue, from the behavioural component, which is genuinely uncertain in published data. That distinction matters because it tells the reader which part of the number is anchored in the stock and which part is a scenario.

The retail result is particularly useful for policy communication. It would be tempting to say that households accelerate pass-through by fleeing old retail debt when market rates rise. The Portuguese episode points the other way: households subscribed when the certificate return became attractive. The fiscal problem was not that existing holders had to be replaced at a higher rate; it was that new retail funding arrived on terms tied to the new rate environment. That is still pass-through, but the language is different and the policy lever is different.

The final implication concerns publication practice. If gross subscriptions and redemptions were published in a stable machine-readable series, the behavioural block could be estimated directly rather than treated as a bound. The paper therefore has a data-policy conclusion as well as a modelling one: routine publication of gross retail flows would materially improve public understanding of interest-cost pass-through.

11 Backtest

We freeze the specification, estimate to a cut date, and predict forward using only information available at that date, feeding in realised issuance yields so that kernel error is isolated from yield-path error. The realised effective rate is imported from the companion paper.

Table 6: Out-of-sample backtest errors

Cut year	Model	Mean abs. error (bps)	Worst error (bps)	<i>N</i>
2014	estimated_kernel	52.44	130.95	11
2014	wam_benchmark	55.69	129.07	11
2014	immediate_full_pass_through	95.09	178.70	11
2014	random_walk	120.75	197.39	11
2018	wam_benchmark	45.16	90.60	7
2018	estimated_kernel	47.81	87.74	7
2018	random_walk	63.62	104.35	7
2018	immediate_full_pass_through	119.67	178.70	7
2021	estimated_kernel	14.24	24.83	4
2021	wam_benchmark	14.66	22.85	4
2021	random_walk	24.52	32.45	4
2021	immediate_full_pass_through	81.05	115.72	4

Errors are basis points on the realised average debt rate. Source: author calculations.

Table 7: Model comparison across backtest cut years

Model	Mean abs. error (bps)	Worst error (bps)	Wins	Cuts
estimated_kernel	38.17	130.95	2	3
wam_benchmark	38.50	129.07	1	3
random_walk	69.63	197.39	0	3
immediate_full_pass_through	98.60	178.70	0	3

Wins count cut years in which the model has the lowest mean absolute error. Source: author calculations.

The estimated kernel does not beat the benchmark in a way that would justify forecasting claims. It wins narrowly at two cut dates, loses at one, and every margin is small relative to the error levels. The 2021 cut places the entire tightening episode out of sample, which is where a behaviourally responsive kernel should have won most clearly. The honest reading is that the kernel is useful for decomposing timing, not for promising lower forecast errors in a short annual backtest.

This is consistent with the rest of the paper rather than in tension with it. The compositional bias is concentrated in the first year, and a backtest over the available annual observations has little power to detect it against effective-rate volatility. The backtest is still necessary because it prevents the kernel from being sold as a superior forecasting model when its strongest evidence is a measurement decomposition.

12 Discussion and limitations

The contribution is a measurement result: the standard pass-through proxy is biased, the bias is signable and quantified, and its mechanism is compositional. Anyone using weighted average maturity to time a rate shock through a budget should adjust for the share of the stock that reprices without exiting. This is not a refinement for the sake of refinement. A first-year budget exercise is exactly where floating and indexed liabilities matter, and that is exactly where a maturity-only proxy is least able to see them.

The behavioural thesis is not supported in the strong form. The channel is visible in the raw data — a 151 percent inflow and an abrupt stop at a dated policy change — but it is not identifiable at the precision published sources allow, principally because gross retail flows are not published and the contractual remuneration rate is not machine-readable. That is a useful negative result. It prevents the paper from turning an attractive Portuguese institutional episode into a broad claim about household portfolio elasticities.

The limitations are therefore part of the argument rather than a disclaimer at the end. The contractual track rests on a stylised retirement profile in the absence of a published schedule; the unit of observation is a class-month, not a holder; and the sample contains a single large tightening episode. The model also treats reset instruments as repricing within the first year rather than modelling their exact reset dates. That assumption is appropriate for annual fiscal pass-through, but it is too coarse for monthly cash-flow management.

13 Conclusion

Weighted average maturity understates the speed at which a rate shock reaches a sovereign interest bill, by 10.90 percentage points of the stock at one year for Portugal, because a maturity-calibrated hazard cannot represent instruments that reprice without exiting. A behavioural channel exists and runs opposite to the intuitive direction, but the published data does not identify it.

The practical recommendation is simple. Start from weighted average maturity only for the part of the stock whose repricing is tied to maturity. Put floating-rate and indexed debt on a reset track, put retail instruments on a separate subscription and redemption track where gross flows exist, and then translate euro interest effects into burden ratios under explicit GDP paths. That workflow is more demanding than the one-number WAM proxy, but it prevents the proxy from assigning the wrong timing to instruments whose repricing clock is not a maturity clock.

Data availability

All inputs are public. Instrument-class stocks and portfolio indicators come from IGCP historical series; competing-return covariates from the ECB Data Portal; aggregate debt and interest from Eurostat. The aggregate series are those of the companion interest-burden study [1], used unchanged.

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